

The maximum length of a chess game under the 2023 FIDE Laws

Junyeop Yim

Department of Applied Mathematics, Kongju National University
junyeobe0315@smail.kongju.ac.kr

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Abstract

The FIDE Laws of Chess effective from 1 January 2023 terminate a game automatically upon fivefold repetition or after 75 consecutive moves by each player without a pawn move or a capture. Legal games of 17,697 plies were previously known, and a scheduling analysis of the pawn moves and captures gave an arithmetic upper bound of 17,699 plies, but games of 17,698 or 17,699 plies were not excluded. We close this gap. Partitioning play at pawn moves and captures yields at most 118 segments. Each segment has length at most 150 plies, and each change in the colour of successive segment endpoints reduces this bound by one ply. We prove that a game with all 118 segments must incur at least three such changes. Hence every game has at most $150 \cdot 118 - 3 = 17,697$ plies, and the known constructions are optimal. Moreover, in every maximum-length game, all sixteen pawns make six one-rank moves and promote.

1 Introduction

How long can a game of chess be under the mandatory termination rules? Before 2014, the answer depended on whether a player chose to claim a draw. Threefold repetition and the fifty-move rule were claim-based, so cooperating players could decline to claim and continue indefinitely; classical constructions of unending play were given by Euwe [2] and by Morse and Hedlund [7]. On 1 July 2014, FIDE introduced automatic draws after fivefold repetition and after 75 moves by each player without a pawn move or a capture, making the extremal problem finite [3]. This paper uses the version of the Laws effective from 1 January 2023 [4], which remains in force at the time of writing.

The bookkeeping behind the classical problem was developed by chess problemists. Bondorff, Fabel, and Riihimaa [1] summarise the count of pawn moves and captures that can postpone a draw under the fifty-move rule. Heimo [5] observed that replacing 50 by 75 in this count gives 17,697 plies. After automatic termination was introduced, Labelle [6] and Murphy [8] exhibited legal games of this length. Murphy’s critical-move scheduling analysis produced an arithmetic upper bound of 17,699 plies and left open whether endpoint parity forces a deficit of at least three plies in every game with 118 segments [8, Secs. 2.1 and 3]. We prove that it does, thereby closing the two-ply gap.

Throughout, we disregard time forfeiture and optional endings such as resignation, draws by agreement, and claimable draws. Play is continued from the standard initial position until the first mandatory termination condition under the 2023 Laws applies.

Theorem 1.1. *Under the FIDE Laws of Chess effective from 1 January 2023, every legal move sequence from the standard initial position, continued until the first mandatory termination condition applies, has length at most 17,697 plies. This bound is attained.*

Equivalently, a maximum-length game consists of 8,848 complete move pairs followed by one additional White ply.

The proof of the upper bound reduces to three estimates. Write L for the game length. A *critical move* is a pawn move or a capture, and critical moves divide a game into segments. If K is the number of segments and S counts changes in the colour of successive segment endpoints, with the start of play assigned Black parity, then

$$L \leq 150K - S, \quad K \leq 118, \quad K = 118 \implies S \geq 3. \quad (1)$$

The first estimate follows from the seventy-five-move rule and the parity of segment endpoints. The second couples the number of pawn moves to the number of pawn captures and then counts the capturable initial pieces. Equality in the second estimate forces all sixteen pawns to make six moves and at least 29 captures to occur. These conditions exclude every critical-move block pattern with at most three blocks and yield the third estimate.

Sections 2–4 establish the three estimates in (1). Section 5 establishes sharpness, completes the proof, and records the structure forced at equality.

2 Rules, conventions, and the segment decomposition

2.1 Rule set and chess conventions

For the purposes of this paper, a *game* is a sequence of legal moves from the standard initial position, continued until the first mandatory termination condition in the 2023 FIDE Laws applies. Write L for its length in *plies*, where one ply is one move by one player; White plays the odd-numbered plies. The mandatory conditions relevant to the statement are as follows.

1. Checkmate (Article 5.1.1), stalemate (Article 5.2.1), and a dead position (Article 5.2.2) terminate the game immediately. Checkmate and stalemate are positions in which the player to move has no legal move, the king of that player being under attack in the case of checkmate and not under attack in the case of stalemate. A dead position is one from which neither player can checkmate by any possible sequence of legal moves.
2. Fivefold repetition (Article 9.6.1) terminates the game: play ends when the same position has appeared at least five times. For this purpose, positions are considered the same if and only if the same player has the move, pieces of the same kind and colour occupy the same squares, and the possible moves of all pieces of both players are the same. In particular, a change in the availability of a legal en passant capture or in castling rights makes the positions different (Article 9.2).

3. The seventy-five-move rule (Article 9.6.2) terminates the game after 75 consecutive moves by each player, equivalently 150 consecutive plies, without a pawn move or a capture. If the final ply delivers checkmate, checkmate takes precedence.

We disregard time forfeiture, resignation, draws by agreement, and claimable draws, since none can increase the number of plies. The upper-bound argument uses Article 9.6.2 throughout and Article 5.2.2 only in Lemma 3.6; the other mandatory termination rules can only shorten a game. Apart from these rules, the proof uses only alternation of play and the elementary facts about the pieces recorded below.

We use the standard coordinates of a chessboard: the files are $a, b, \dots, h$ and the ranks are $1, 2, \dots, 8$. Each side begins with sixteen pieces: a king, a queen, two rooks, two bishops, two knights, and eight pawns (Articles 2.2–2.3). Throughout the proof, *piece* includes a pawn. A *capture* is a move to a square occupied by an opposing piece; the captured piece is removed from the board as part of the same move and takes no further part in the game (Article 3.1.1). The en passant capture, described below, is the one exception, in which the captured piece does not stand on the destination square.

White pawns begin on rank 2 and Black pawns on rank 7, one pawn of each colour on each file; White pawns move towards rank 8 and Black pawns towards rank 1. We call ranks 2 and 7 their respective home ranks, and ranks 8 and 1 their promotion ranks. A pawn advances one rank to an unoccupied square or, on its first move from its home rank, two ranks provided that both squares ahead of it are unoccupied (Articles 3.7.1–3.7.2). It captures one square diagonally forward onto a square occupied by an opposing piece (Article 3.7.3). In the en passant capture (Articles 3.7.3.1–3.7.3.2), immediately after an opposing pawn has advanced two ranks from its home rank, a pawn may capture it as though it had advanced only one rank, moving diagonally forward onto the empty square that the opposing pawn passed over. While it remains a pawn, it changes file only by making a capture. Upon reaching the promotion rank a pawn must, as part of the same move, be exchanged for a new queen, rook, bishop, or knight of its own colour (Article 3.7.3.3); after promotion it is no longer a pawn.

A piece *attacks* a square if the movement rules would allow it to capture an opposing piece standing there, even when such a move is otherwise illegal (Articles 3.1.2–3.1.3); a king is *under attack* when an opposing piece attacks the square it occupies (Article 3.9.1). The king moves one square in any direction (Article 3.8.1), so it attacks precisely the adjacent squares.

Kings are never captured, since capturing a king is not allowed (Article 1.4.1); hence each side has fifteen capturable initial pieces: eight pawns and seven non-pawn pieces. We follow each of the 32 initial pieces through the game, retaining its identity after promotion. Thus a promoted pawn remains one of the eight pieces that began as pawns, and promotion creates no additional capturable identity.

2.2 Critical moves and segments

Definition 2.1 (Critical and noncritical moves). A *critical move* is a pawn move or a capture. A *noncritical move* is any other move. Thus an en passant capture is critical for both reasons,

promotion is critical because it is a pawn move, and castling¹ is noncritical.

Critical moves are precisely the moves that reset the counter in Article 9.6.2.

Definition 2.2 (Segments and endpoints). Let

$$t_1 < t_2 < \cdots < t_m$$

be the plies on which the critical moves of a game occur. Set $T = 1$ if at least one noncritical ply follows the last critical move, or if the game contains no critical move, and set $T = 0$ otherwise. Define

$$e_0 = 0, \quad e_i = t_i \quad (1 \leq i \leq m),$$

and, when $T = 1$, define $e_{m+1} = L$. Put $K = m + T$. For $1 \leq i \leq K$, the interval of plies

$$(e_{i-1}, e_i] \cap \mathbb{Z} = \{e_{i-1} + 1, \dots, e_i\}$$

is *segment i* . When $T = 1$, the final segment is called the *terminal noncritical segment*.

For $i \geq 1$, let $\text{col}(e_i)$ be the colour of the player who moves on ply e_i : White when e_i is odd and Black when it is even. We assign the virtual endpoint $e_0 = 0$ the colour Black. This convention treats the start of the game as having the parity of a preceding Black ply, so that the parity relation between segment length and endpoint colours in Lemma 2.4 applies to the first segment as well: a first segment ending on a White ply has odd length.

Definition 2.3 (Switch count). The *switch count* is

$$S = \#\{i \in \{1, \dots, K\} : \text{col}(e_i) \neq \text{col}(e_{i-1})\}.$$

For example, the following endpoint parities give three segments and one switch:

$$0_B \xrightarrow{150} 150_B \xrightarrow{149} 299_W \xrightarrow{150} 449_W.$$

The total length is therefore $150 \cdot 3 - 1 = 449$.

Lemma 2.4 (Segment decomposition). *Every game satisfies*

$$L = 150K - S - \sum_{i=1}^K \delta_i, \quad \delta_i \in \mathbb{Z}_{\geq 0}. \quad (2)$$

Here δ_i is the shortfall of segment i from length 150 when its endpoints have the same colour, and from length 149 when they have different colours. In particular,

$$L \leq 150K - S.$$

Proof. Fix a segment with endpoints e_{i-1} and e_i . Every ply in the segment except possibly its last is noncritical; in a terminal noncritical segment, the last ply is noncritical as well. If the segment had length at least 151, then its first 150 plies would be noncritical, and Article 9.6.2 would terminate the game at ply $e_{i-1} + 150$. Hence every segment has length at most 150.

Because plies alternate colours, $e_i - e_{i-1}$ is even exactly when the two endpoints have the same colour. A segment whose endpoints have different colours therefore has odd length and hence length at most 149. Subtracting each segment length from the corresponding bound defines $\delta_i \in \mathbb{Z}_{\geq 0}$. Summing over the segments gives (2). $\square$

¹Castling (Article 3.8.2) is a combined move of the king and one rook of the same colour; it counts as a single move and captures nothing.

Let P be the number of pawn moves, C the number of captures, and C_p the number of captures made by pawns. The number of critical moves is $P + C - C_p$, and therefore

$$K = (P - C_p) + (C + T). \quad (3)$$

3 The 118-segment bound

Lemma 3.1 (Pawn moves). *Each pawn makes at most six pawn moves. Consequently $P \leq 96$. If a pawn makes six pawn moves, every one of them advances it by exactly one rank, and the sixth move promotes it.*

Proof. The promotion rank is six rank-steps from the home rank, and every pawn move advances the pawn by at least one rank towards promotion. Equality therefore requires six one-rank moves, the last of which promotes the pawn. $\square$

3.1 Pawn moves and pawn captures

Definition 3.2 (Origin pairs). For each file, the White pawn and Black pawn that begin on that file form its *origin pair*. An origin pair is *resolved* if at least one of its members makes a capture while still a pawn, and *unresolved* otherwise. A capture made on the promotion move counts as a capture by a pawn; a capture made by the promoted piece on a later move does not. Let f be the number of resolved origin pairs.

Lemma 3.3 (Unresolved origin pair). *The two pawns in an unresolved origin pair make at most ten pawn moves in total.*

Proof. While both pawns remain on the board as pawns, neither changes file, so they remain on their common origin file with the White pawn strictly below the Black pawn: straight pawn advances require unoccupied squares, so the two pawns can neither meet nor pass one another. If they have advanced a and b ranks, respectively, then

$$2 + a < 7 - b,$$

and hence $a + b \leq 4$. Thus the pair makes at most four pawn moves while both members remain on the board. In particular, neither pawn can promote before the other is captured.

If neither pawn is ever captured, their combined total is at most four. Otherwise, the pawn captured first has made at most four moves, while the surviving pawn makes at most six moves over its entire lifetime by Lemma 3.1. The combined total is therefore at most $4 + 6 = 10$. $\square$

Lemma 3.4 (Pawn-capture lower bound). $C_p \geq f$.

Proof. For each resolved origin pair, choose the first capture made by one of its pawns. This move contributes to C_p . Different origin pairs give different moving pawns and hence different moves. $\square$

Proposition 3.5 (Net pawn-move bound). *For every game,*

$$P - C_p \leq 88.$$

Equality implies

$$f = 8, \quad P = 96, \quad C_p = 8.$$

Proof. For each unresolved origin pair, Lemma 3.3 gives at most ten pawn moves. For each resolved origin pair, Lemma 3.1 gives at most twelve. Hence

$$P \leq \min\{96, 10(8 - f) + 12f\} = \min\{96, 80 + 2f\}.$$

Together with $C_p \geq f$, this gives

$$P - C_p \leq \min\{96, 80 + 2f\} - f.$$

For $f \leq 7$, the right-hand side is at most $80 + f \leq 87$; for $f = 8$, it is at most $96 - 8 = 88$. Equality therefore requires $f = 8$, after which the bounds $P \leq 96$ and $C_p \geq 8$ force $P = 96$ and $C_p = 8$. $\square$

3.2 Captures and the terminal noncritical segment

Lemma 3.6 (Capture-terminal bound). $C + T \leq 30$.

Proof. There are 30 capturable initial pieces, so $C \leq 30$. If $C \leq 29$, then $C + T \leq 30$ because $T \leq 1$. If $C = 30$, only the two kings remain after the last capture. A position containing only the two kings is dead,² so by Article 5.2.2 no ply can follow that capture. Hence no terminal noncritical segment exists, $T = 0$, and again $C + T = 30$. $\square$

3.3 The segment bound and its equality case

Theorem 3.7 (Segment bound). *Every game satisfies $K \leq 118$. If $K = 118$, then*

$$f = 8, \quad P = 96, \quad C_p = 8, \quad C + T = 30. \quad (4)$$

In particular, all sixteen pawns make six one-rank moves, promoting on the sixth, and $C \geq 29$.

Proof. By (3), Proposition 3.5, and Lemma 3.6,

$$K = (P - C_p) + (C + T) \leq 88 + 30 = 118.$$

Equality requires equality in both component bounds. Proposition 3.5 gives $f = 8$, $P = 96$, and $C_p = 8$, while Lemma 3.6 gives $C + T = 30$. Since $T \leq 1$, we have $C \geq 29$. Finally, $P = 96$ forces every pawn to attain the equality case of Lemma 3.1. $\square$

4 The three-switch bound

Throughout this section, assume $K = 118$. By Theorem 3.7, every pawn makes six pawn moves and at least 29 captures occur. Since $T \leq 1$, the game has at least 117 critical moves; in particular, the critical-move colour sequence is nonempty.

²With only the kings on the board, neither king can be under attack: the only possible attacker is the opposing king, and the two kings can never stand on adjacent squares, because the move creating such a position would place the moving player's own king under attack, which is illegal (Article 3.9.2). Since checkmate requires the king to be under attack, no sequence of legal moves leads to checkmate.

4.1 Block patterns

Definition 4.1 (Block pattern). For a nonempty finite sequence with entries in $\{B, W\}$, its *block pattern* is obtained by replacing each maximal constant run by a single letter; these maximal runs are the *blocks* of the sequence. The *critical-move colour sequence* is

$$(\text{col}(t_1), \dots, \text{col}(t_m)),$$

and the *endpoint colour sequence* is

$$(\text{col}(e_1), \dots, \text{col}(e_K)).$$

Their block patterns are called the *critical-move block pattern* and the *endpoint block pattern*, respectively.

For example,

$$(B, B, W, W, B) \mapsto BWB.$$

The virtual endpoint e_0 is not part of the endpoint colour sequence; it is used only in the definition of S .

Lemma 4.2 (Endpoint blocks). *The endpoint colour sequence has at least as many blocks as the critical-move colour sequence. If the endpoint colour sequence has r blocks, then*

$$S = \begin{cases} r - 1, & \text{if the first endpoint is Black,} \\ r, & \text{if the first endpoint is White.} \end{cases}$$

Consequently, an endpoint colour sequence with at least four blocks has $S \geq 3$.

Proof. The endpoint colour sequence is the critical-move colour sequence, with one terminal endpoint appended when $T = 1$. Appending an element cannot reduce the number of blocks. The formula for S counts the $r - 1$ changes between consecutive blocks and, when the first endpoint is White, the additional change from the virtual Black endpoint e_0 . $\square$

The possible critical-move block patterns with at most three blocks are

$$B, \quad W, \quad BW, \quad WB, \quad BWB, \quad WBW.$$

We exclude them in turn. For WBW this is more than the inequality $S \geq 3$ requires: its initial White endpoint already incurs a switch against the virtual Black endpoint e_0 , so Lemma 4.2 gives $S \geq 3$ directly. The exclusion nevertheless matters, because the four-block conclusion of Theorem 4.7 is what determines the block pattern of maximum-length games in Corollary 5.2.

4.2 One and two blocks

Proposition 4.3. *The block patterns B and W are impossible when $K = 118$.*

Proof. A pawn move is a critical move of the pawn's side. With only one block, only one side's eight pawns can move, so $P \leq 8 \cdot 6 = 48$, contrary to $P = 96$. $\square$

Proposition 4.4. *The block patterns BW and WB are impossible when $K = 118$.*

Proof. Write the pattern as (X, Y) , so every critical move by X precedes every critical move by Y . Let C_X and C_Y be the respective numbers of captures. Since Y can capture at most the fifteen capturable pieces of X , we have $C_Y \leq 15$, and therefore

$$C_X = C - C_Y \geq 29 - 15 = 14.$$

All these captures occur before the first critical move by Y . Among the captured Y -pieces, at most seven did not begin as pawns, so at least seven Y -pawns are captured before making any pawn move. They cannot move after they are captured. Hence

$$P \leq 96 - 7 \cdot 6 = 54,$$

contrary to $P = 96$. □

4.3 A home-rank obstruction

Lemma 4.5 (Home-rank lemma). *Let X and Y be the two opposing sides. Consider a prefix of a legal game from the standard initial position. Suppose that*

1. *every move made by Y in this prefix is noncritical;*
2. *no move made by X in this prefix captures a Y -pawn.*

Then all eight Y -pawns remain on their home squares, no X -pawn reaches Y 's pawn home rank, and every X -pawn makes at most four pawn moves during the prefix.

Proof. A move by a Y -pawn would be critical, so the first condition implies that no Y -pawn has moved. The second condition implies that none has been captured. Thus every Y -pawn remains on its initial square.

An X -pawn cannot move onto Y 's pawn home rank. A forward move to an occupied square is illegal, while a diagonal move to such a square would capture the Y -pawn, contrary to the second condition. Between an X -pawn's home rank and Y 's pawn home rank there are four ranks on which it may stand. Every pawn move advances it by at least one rank, so it makes at most four pawn moves during the prefix. □

4.4 Three blocks

The remaining case is ruled out by the home-rank obstruction.

Proposition 4.6. *The block patterns BWB and WBW are impossible when $K = 118$.*

Proof. Write the pattern as (X, Y, X) , and let C_X, C_Y be the numbers of captures made by the two sides. Since X can capture at most fifteen Y -pieces,

$$C_Y = C - C_X \geq 29 - 15 = 14.$$

At most seven of the pieces captured by Y did not begin as pawns. Hence at least seven of the captured X -pieces began as pawns.

Let τ be the ply on which Y makes its first critical move, and fix one such piece p , which began as an X -pawn. By Theorem 3.7, p makes six pawn moves before it is captured. Any

critical move by X after τ belongs to the final X -block and therefore occurs after every critical move in the middle Y -block, including the capture of p . Consequently all six pawn moves of p occur before τ .

Consider the game prefix ending immediately before τ . Every move made by Y in this prefix is noncritical. Moreover, X cannot have captured a Y -pawn: such a pawn would have been removed before making any critical move and could not make the six pawn moves required by $P = 96$. Lemma 4.5 therefore applies to this prefix and implies that p makes at most four pawn moves before τ , contradicting the six moves just established. $\square$

Theorem 4.7 (Three-switch bound). *If $K = 118$, then the critical-move colour sequence has at least four blocks and $S \geq 3$.*

Proof. Propositions 4.3, 4.4, and 4.6 exclude all nonempty block patterns with at most three blocks. Hence the critical-move colour sequence has at least four blocks. Lemma 4.2 transfers this lower bound to the endpoint colour sequence and gives $S \geq 3$. $\square$

5 Completion of the proof and the equality case

Proposition 5.1 (Sharpness). *There exists a legal move sequence of 17,697 plies from the standard initial position such that no mandatory termination condition applies before the final ply, and the final ply delivers checkmate.*

Proof. Labelle [6] and Murphy [8] gave explicit legal games of this length. Murphy's move sequence has been verified by replay [9]: the replay checks move legality, fivefold repetition, and the seventy-five-move rule after every ply, and confirms checkmate on the final ply. It includes no decision procedure for dead positions under Article 5.2.2, but no such procedure is needed for this particular game. From every earlier position, the remaining verified suffix is a legal sequence ending in checkmate, so that position is not dead; the existence of the next legal ply also rules out earlier checkmate and stalemate. Hence no mandatory termination condition applies before the final ply, and the lower bound follows. $\square$

Proof of Theorem 1.1. Let K be the number of segments. By Theorem 3.7, $K \leq 118$. If $K \leq 117$, then Lemma 2.4 gives

$$L \leq 150 \cdot 117 = 17,550.$$

If $K = 118$, then Theorem 4.7 gives $S \geq 3$, and therefore

$$L \leq 150 \cdot 118 - 3 = 17,697.$$

This proves the universal upper bound. Proposition 5.1 shows that it is attained. $\square$

Corollary 5.2 (Structure of maximum-length games). *Every game of length 17,697 has*

$$K = 118, \quad S = 3, \quad \delta_i = 0 \quad (1 \leq i \leq 118),$$

and

$$f = 8, \quad P = 96, \quad C_p = 8, \quad C + T = 30.$$

Thus all sixteen pawns make six one-rank moves, promoting on the sixth; at least 29 captures occur; and every segment has the maximum length permitted by its endpoint parity. The critical-move and endpoint block patterns are both BWBW.

Proof. A game with $K \leq 117$ has length at most 17,550, so a maximum-length game has $K = 118$. Combining Theorem 4.7 with (2) gives

$$17,697 = 150 \cdot 118 - S - \sum_{i=1}^{118} \delta_i \leq 150 \cdot 118 - 3.$$

Equality holds throughout. Hence $S = 3$ and every $\delta_i = 0$. The equality statement in Theorem 3.7 gives the remaining numerical assertions and the behaviour of the pawns.

The critical-move colour sequence has at least four blocks by Theorem 4.7. Each transition between consecutive blocks of this sequence is a change of colour between consecutive endpoints and contributes one switch, so $S = 3$ permits at most four blocks. Thus there are exactly four. If the first block were White, the comparison with the virtual Black endpoint would contribute a fourth switch. Hence the critical-move block pattern is BWBW. Appending a terminal endpoint cannot create a new block without increasing S , so the endpoint block pattern is the same. $\square$

Data and code availability

The companion archive [9] contains Murphy’s 17,697-ply move sequence, the two separately implemented replay verifiers that perform the checks described in the proof of Proposition 5.1, and ancillary computational cross-checks. None of these computations enters the upper-bound proof. Verification instructions accompany the archive; the development version and detailed verification notes are maintained in the [project repository on GitHub](#).

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